CHAT SERVER AND CLIENT USING TCP SOCKET PROGRAMMING IN C

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Overview:

This is a simple chat server built in C using TCP Sockets. It lets multiple clients connect to a single server at the same time so they can talk to each other. You can broadcast a message to everyone at once or have a private chat with just one person.

Prerequisites:

- ➢ A GCC Compiler
- ➢ A Linux-based Environment (or macOS)

File Structure:

- ➢ server.c : The code for the server-side logic.
- ➢ client.c: The code for client-side logic.

Instructions for running the codes:

- gcc server.c -o server -lpthread
- ./server 8080
- gcc client.c -o client -lpthread
- ./client 127.0.0.1 8080
- The above are the basic requirements you can add more clients.

General Format for running the Codes:

- gcc server.c -o server -lpthread
- ./server <port_no_of_server>
- gcc client.c -o client -lpthread
- ./client <ip_address_of_server> <port_no_of_server>

Primary Functionalities of the Code:

Choice Number	Functionality	Brief
2	Direct Message	Two clients can chat via the server
3	Broadcast	A Client can Broadcast to all clients
12	Exit	Client can disconnect from the chat

Images of running server and clients:

I am running the codes with three clients and 1 server.
Running server first

```
harsha@Harshas-MacBook-Air my_project % gcc server.c -o a -lpthread
harsha@Harshas-MacBook-Air my_project % ./a 8080
[20:59:16] Server is Listening....!
```

Then let us run the client to check whether the code allows the duplicate names

```
harsha@Harshas-MacBook-Air my_project % gcc client.c -o a -lpthread
harsha@Harshas-MacBook-Air my_project % ./a 127.0.0.1 8080
[21:01:44] Connected to server!
[21:01:44] Enter username: alice
[21:01:48] Username accepted!
```

```
harsha@Harshas-MacBook-Air my_project % gcc client.c -o a -lpthread
harsha@Harshas-MacBook-Air my_project % ./a 127.0.0.1 8080
[21:03:28] Connected to server!
[21:03:28] Enter username: bob
[21:03:32] Username accepted!
```

```
harsha@Harshas-MacBook-Air my_project % gcc client.c -o a -lpthread
harsha@Harshas-MacBook-Air my_project % ./a 127.0.0.1 8080
[21:04:15] Connected to server!
[21:04:15] Enter username: alice
[21:04:18] Username taken, try again
[21:04:18] Enter username: kumar
[21:05:25] Username accepted!
```

Now I will paste the choice/Menu here for the functionalities for the client

```
[21:05:25]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

Now let us the test the choice 1 (List active users)

```
Enter choice: 1

[21:10:21]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [21:10:21] Users:
alice
bob
kumar
```

Now let us test choice 2 (Direct Message)

Alice's screen:

```
2
[21:14:14] Enter receiver: bob
[21:14:17] Enter message: hi bob

[21:14:27]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

Bob's screen:

```
[21:14:27] [21:14:27] [DM from alice]: hi bob
```

Now let us test choice 3 (Broadcasting all clients)

```
3
[21:19:26] Enter message: hi everyone

[21:19:31]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

```
[21:14:27]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [21:19:31] [21:19:31] bob: hi everyone
```

```
[21:19:31] [21:19:31] bob: hi everyone
```

Now let us test choice 6&7 (Joining and Creation of group)

Alice's screen

```
6
[21:29:01] Enter group name: g1

[21:29:05]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [21:29:18] [21:29:18] [bob to group g1]: [21:29:18] [Added in the
group g1]
```

kumar's screen

```
[21:29:05] [21:29:05] [group is created g1]
```

Bob's screen after joining group g1

```
7
[21:29:15] Enter group name: g1

[21:29:18]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

Now let us test choice 9 (Messaging in a group)

Alice's screen

```
9
[21:34:14] Enter group name: g1
[21:34:15] Enter message: hello group

[21:34:21]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

Bob's screen

```
[21:34:21] [21:34:21] [alice to group g1]: hello group
```

Now let us test choice 5 (Listing the groups)

```
5

[21:41:17]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [21:41:17] Groups:
g1
```

Now let us test the choice 8 (renaming the group)

```
8
[21:39:36] Old group name: g1
[21:39:38] New group name: newg
```

```
Enter choice: 5

[21:42:45]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [21:42:45] Groups:
newg
```

Now let us the test the choice 9 (Messaging the group even if not in the group)

kumar's screen

```
9
[21:44:06] Enter group name: newg
[21:44:10] Enter message: Hi everyone

[21:44:18]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

Alice's screen(already in the group newg)

```
[21:44:18] [21:44:18] [kumar to group newg]: Hi everyone
```

Bob's screen(already in the group newg)

```
[21:44:18] [21:44:18] [kumar to group newg]: Hi everyone
```

Now let us test the choice 10(leaving the group)

```
Enter choice: 10
[23:18:14] Enter group name: newg

[23:18:15]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [23:18:15] [23:18:15] [bob left from group newg]
```

Now let us test the choice 11(deleting the group)

```
11
[21:55:20] Enter group name: newg

[21:55:22]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice:
```

```
5

[21:55:28]
==== MENU ====
1 -> List users
2 -> Direct Message
3 -> Broadcast
4 -> Rename User
5 -> List Groups
6 -> Create Group
7 -> Join Group
8 -> Rename Group
9-> Message in group
10-> Leave Group
11 -> Delete Group
12 -> Exit
Enter choice: [21:55:28] Groups:
```

Now let us test the choice 4(renaming of a user)

kumar's screen

```
Enter choice: 4
[21:58:12] Enter new username: Ram
```

Name change is broadcasted to all the active clients

```
[21:59:01] [21:59:01] Name of kumar is changed to Ram
```

Now,testing for an invalid choice

```
99
[22:00:59] Invalid choice
```

Now testing the choice 12 (Disconnecting from the server)

```
12
[22:00:51] Disconnected
harsha@Harshas-MacBook-Air my_project %
```

Note:

In the last, you have to press the Ctrl+C button to exit the server as the server must run all the time when the clients are active.

Additional functionalities of the Code:

Choice Number	Functionality	Brief
1	List Users	Displays all active clients
4	Rename user	Allows a client to change their username
5	List Groups	Shows all the chat groups that are available in the server
6	Create group	Allows a client to create a new group
7	Join group	Allows a client to join any existing group
8	Renaming group	Allows any client to rename the group
9	Message in group	Allows any client to message in the group
10	Leave group	Allows the client to exit a group there are in
11	Delete group	Removes the group from the server